Year 9 Science – Booklet 3: Physics

Space Physics — ANSWERS

4.1 The night sky

In-text questions

A. The Moon.

B. Mercury, Venus, Mars, Jupiter and Saturn.

C. A comet appears as a large, bright, often hazy object with a glowing tail, moving slowly across the sky over many nights as it orbits the Sun. A meteor appears as a brief, fast streak of light ("shooting star") lasting only a fraction of a second, caused by a small piece of dust or rock burning up in the Earth's atmosphere.

D. A galaxy is a huge collection of billions of stars (held together by gravity) – our own galaxy is the Milky Way.

Summary Questions

1. Choose the correct bold word (4 marks)

There are thousands of satellites in orbit around the **Earth**. The Moon is a natural satellite of the **Earth**. Comets are huge snowballs that orbit the **Sun**. Planets orbit the **Sun**.

2. Meteor vs meteorite (2 marks)

A meteor is a piece of dust or rock that burns up as it moves through the Earth's atmosphere, producing a streak of light. A meteorite is a meteor that survives the journey through the atmosphere and actually reaches the ground.

3. What could the (unmoving) dots of light be? (2 marks)

Stars in our own galaxy (the Milky Way), or other galaxies (e.g. Andromeda).

4. Compare the time light takes to reach us from different objects (6 marks QWC)

Light travel time increases hugely with distance: from the Moon it takes just over a second; from the Sun about 8 minutes; from Saturn about 1.5 hours; from our next-nearest star (after the Sun) over 4 years, since it is roughly 4 light-years away; and from the Andromeda galaxy, our nearest large galaxy, about 2 million years. So travel time jumps by many orders of magnitude as you move from objects within the Solar System, to nearby stars, to other galaxies.

4.2 The Solar System

In-text questions

A. Eight.

B. Mercury, Mars, Venus, Earth, Neptune, Uranus, Saturn, Jupiter (smallest to largest, using the diameters in the table).

Summary Questions

1. Copy and complete the sentences (5 marks)

There are **four** inner and **four** outer planets in the Solar System. The band of dust and rocks between Jupiter and Mars is called the **asteroid belt**. Pluto is a **dwarf** planet. Scientists think that most comets come from a region outside the Solar System called the **Oort Cloud**.

2. One similarity and one difference, inner vs outer planets (2 marks)

Similarity: both types orbit the Sun. Difference: inner planets are small, rocky (terrestrial) and hotter, while outer planets are much bigger, made mainly of gas, and colder.

3. Compare planets and asteroids (2 marks)

Both orbit the Sun and can be rocky, but planets are much larger, roughly spherical (rounded by their own gravity), and may have moons, whereas asteroids are small, irregularly-shaped lumps of rock, mostly confined to the asteroid belt between Mars and Jupiter.

4. Link between distance from the Sun and temperature; why is Venus the odd one out? (6 marks QWC)

In general, the further a planet is from the Sun, the less solar energy it receives, so the colder its average temperature – this is why the outer gas giants are far colder than the inner rocky planets. Venus is the odd one out because, despite being further from the Sun than Mercury, it is hotter (about 465°C, hotter than Mercury's daytime maximum). This is because Venus has a thick carbon dioxide atmosphere that traps energy from the Sun (a strong greenhouse effect), heating the planet far more than its distance from the Sun alone would suggest.

4.3 The Earth

In-text questions

A. Take a long-exposure photograph of the night sky over several hours – the stars appear to trace circular trails around a fixed point, showing that the Earth (and the camera on it) is rotating while the stars themselves stay still.

B. East.

Summary Questions

1. Choose the correct bold word (7 marks)

You see the Sun rise in the **east** and set in the **west** because the Earth **spins**. A **year** lasts approximately 365 days. This is the time that it takes the Earth to **orbit the Sun**. The days are **longer** in the summer and the Sun is **higher** in the sky at noon.

2. a) Why is it hotter in summer? (2 marks)

In summer the hemisphere is tilted towards the Sun, so the Sun's rays are more concentrated over a smaller area (rather than spread out) and the days are longer, so more solar energy is received overall.

2. b) Why is a fence post's shadow longer in winter? (1 mark)

In winter the Sun is lower in the sky at noon, so light hits the post at a shallower angle, casting a longer shadow.

3. What would you experience if the Earth's axis was not tilted? (6 marks QWC)

There would be no seasons: every place on Earth would receive roughly the same amount and intensity of sunlight all year round, day and night would both stay close to 12 hours everywhere throughout the year, and the average temperature at any given location would remain roughly constant instead of varying between a hot summer and cold winter.

4.4 The Moon

In-text questions

A. Full, gibbous, third quarter, crescent, new, crescent, first quarter, gibbous (back to full).

B. Half of the Moon's surface is always lit by the Sun (as always) – but at new moon that lit half faces entirely away from Earth, so none of the illuminated part is visible from Earth.

In-text question (p.155)

C. The umbra.

Summary Questions

1. Copy and complete the sentences (6 marks)

You see a **full** moon when the Sun lights up the whole of the side that you can see. When the side of the Moon that you can see is in shadow you see a **new** moon. A solar eclipse happens when the **Moon** comes between the Sun and the **Earth**. A lunar eclipse happens when the **Earth** comes between the Sun and the **Moon**.

2. Why can you see an eclipse on some planets but not others? (1 mark)

An eclipse of this kind needs a moon to pass between its planet and the Sun (or vice versa), so only planets that have at least one moon can experience one – planets with no moons (e.g. Mercury and Venus) cannot.

3. Torch, beach ball and tennis ball demonstration (6 marks QWC)

Let the torch represent the Sun, the beach ball represent the Earth, and the tennis ball represent the Moon. Shine the torch steadily at the beach ball. For a solar eclipse, place the tennis ball between the torch and the beach ball, so its small shadow falls on part of the beach ball – showing the Moon coming between the Sun and Earth. For a lunar eclipse, move the tennis ball to the far side, so the beach ball sits between the torch and the tennis ball – its much larger shadow falls on the tennis ball, showing the Earth coming between the Sun and the Moon.

P4.1 The night sky (workbook)

A. Fill in the gaps

We can see many different objects in the night sky without a telescope. The nearest are artificial **satellites** that **orbit** the Earth. The **Moon** is the Earth's only **natural** satellite. We can also see five **planets**: Mercury, Venus, **Mars**, Jupiter, and Saturn. Like Earth, they orbit the **Sun** and make up part of the **Solar System**. We may also be lucky enough to see meteors or **comets** in the night sky. The dots of light we see are **stars** in our own **galaxy**, which is called the **Milky Way**. It is just one of billions of galaxies that make up the **Universe**.

B. Describe and define

a. Meteor – a small piece of dust or rock that burns up as it moves through the Earth's atmosphere, producing a streak of light.

b. Solar system – the Sun together with all the objects that orbit it (planets, dwarf planets, moons, asteroids and comets).

c. Galaxy – a huge collection of billions of stars (and other matter) held together by gravity.

d. Star – a huge ball of burning gas that gives out its own light, powered by nuclear reactions at its centre.

C. Match each object with its distance from Earth

Proxima Centauri → **4 light-years** • Moon → **1 light-second** • Andromeda → **2 million light-years** • Sun → **8 light-minutes**

D. Structure of the Universe, smallest to largest

Moons are the smallest objects here; they orbit planets. Planets (and asteroids, small rocky bodies) orbit a star such as the Sun. The Sun, with everything orbiting it, makes up our Solar System. Our

Solar System is just a tiny part of a much larger collection of billions of stars called the **Milky Way** (our galaxy). The Milky Way is in turn just one of billions of galaxies that together make up the **Universe**, the largest scale of all.

P4.2 The Solar System (workbook)

A. Fill in the gaps

At the centre of the **Solar** System is the **Sun**. It is orbited by four inner **planets** and four outer **planets**. Each orbit is a squashed circle shape called an **ellipse**. The inner planets are Mercury, **Venus**, Earth, and **Mars**. They are all terrestrial planets; they are made of **rock**. The outer planets are Jupiter, Saturn, Uranus, and Neptune. They are called **gas** giants; they are made mainly of **hydrogen and helium** and are very cold and much bigger than the inner planets. Many planets have **moons** that orbit them. Between the inner and outer planets there is an **asteroid** belt made up of thousands of pieces of rock. Pluto used to be called a planet but is now called a **dwarf** planet.

B. Match object with description

asteroid belt → made up of small rocky bodies, between Mars and Jupiter
Sun → in the centre of the Solar System
a planet → orbits Sun, may be rocky or gas giant
a moon → orbits a planet
a comet → made up of ice and dust; sometimes orbits close to the Sun, sometimes far

C. a) Properties/features linked to place in Solar System

The closer a planet is to the Sun, the hotter, smaller and rockier it tends to be (the inner planets are small, dense, terrestrial worlds); the further out, the colder and larger/more gas-rich it tends to be (the outer planets are huge gas giants), and outer planets tend to have more moons and rings.

C. b) Predict Neptune's features (beyond Uranus, 4500 million km from Sun)

Following the trend in the table, Neptune would be expected to be a gas giant made mainly of hydrogen and helium, very cold (colder than Uranus's –200°C, so roughly –210 to –220°C), similar in size to Uranus (tens of thousands of km diameter), and to have several moons and rings, like the other outer planets.

D. Two similarities and two differences: asteroid vs planet

Similarity 1: Both orbit the Sun.
Similarity 2: Both can be made of rock.
Difference 1: Planets are much larger and roughly spherical (rounded by their own gravity); asteroids are small and irregularly shaped.
Difference 2: Planets follow their own individual orbits around the Sun; most asteroids are confined to the asteroid belt between Mars and Jupiter.

P4.3 The Earth (workbook)

A. Fill in the gaps

Once each **day**, the Earth **spins** on its axis, causing the Sun's apparent motion through the sky and giving us day and **night**. Once each **year**, the Earth **orbits** around the Sun. The differences between summer and winter in the UK are mainly caused by the **tilt** of the Earth's axis. In summer, the Northern Hemisphere tilts **towards** the Sun, which means that each bit of ground receives **more** of the Sun's rays; days are **longer**, and the Sun appears to rise **higher** into the sky at noon.

B. Match object with explanation of apparent movement

the Moon → orbits the Earth once a month
the Sun → the Earth spins on its axis once every 24 hours, so different regions face the Sun at

different times of day
stars → the Earth orbits the Sun once a year, so the night-side of the Earth faces different constellations during the year

C. a) Predict how temperatures compare across the cities in summer

Cities closer to the equator (lower latitude, e.g. Accra) would be expected to be warmer on average than cities further away (higher latitude, e.g. Iqaluit), because locations nearer the equator receive more direct, concentrated sunlight all year, whereas higher-latitude locations receive more spread-out, less intense sunlight.

C. b) Predict daylight hours in Accra and Iqaluit on the longest day

Following the trend that daylight hours increase with latitude (34°→14.4 h, 50°→16.4 h, 57°→17.9 h):
Accra (5°, very close to the equator) ≈ **12 hours** (near-equal day and night all year at the equator).
Iqaluit (63°, higher than Aberdeen) ≈ **19–20 hours** (within the Arctic Circle, summer days are extremely long).

D. Seasons in the UK if the axis was not tilted

There would be no real seasons – the amount and intensity of sunlight reaching the UK would stay roughly the same all year, day length would remain close to 12 hours throughout the year, and temperatures would stay fairly constant instead of varying between warm summers and cold winters.

P4.4 The Moon (workbook)

A. Fill in the gaps

Half of the Moon is lit by the **Sun** at all times. As it orbits the **Earth**, we see different amounts of it illuminated by the Sun, causing the **phases** of the Moon. When the side facing the Earth is in shadow, we call it a **new** moon. Later, when we see the whole of the side lit by the Sun, we call it a **full** moon. This cycle occurs once each lunar **month**. If the Moon is between the **Sun** and the **Earth**, a shadow called the **umbra** occurs on Earth, where the light from the Sun is totally blocked and there will be a total **solar** eclipse. The **penumbra** occurs where only part of the Sun's light is blocked and a partial **solar** eclipse occurs. If the Earth comes between the Sun and the **Moon** and blocks the Sun's light from reaching the Moon, a **lunar** eclipse can occur.

B. Moon-phase diagram positions 1–8

Position	1	2	3	4	5	6	7	8
Phase (letter)	C	E	F	A	B	D	H	G
Phase name	new	crescent	first qtr	gibbous	full	gibbous	third qtr	crescent

(Letters follow the key given on the page: A/D = gibbous, B = full, C = new, E/G = crescent, F = first quarter, H = third quarter, applied to the standard new→crescent→first quarter→gibbous→full→gibbous→third quarter→crescent cycle.)

C. 28-day lunar month, day 1 = new moon – predict phase for each day

Day of lunar month	8	15	22	26	29
Phase of the Moon	first quarter	full	third quarter	waning crescent	new

D. a) How do total solar/lunar eclipses happen?

i). Total solar eclipse: the Moon passes directly between the Sun and Earth, so its umbra (deep shadow) falls on part of Earth's surface, completely blocking the Sun's light there.

ii). Total lunar eclipse: the Earth passes directly between the Sun and the Moon, so the Earth's shadow covers the Moon completely, blocking sunlight from reaching it.

D. b) Which phase is the Moon in for each eclipse?

i). Total solar eclipse – new moon.

ii). Total lunar eclipse – full moon.

E. Why can some planets have eclipses but not others?

An eclipse of this kind needs a moon to pass between its planet and the Sun (or the planet to pass between the Sun and its moon), so only planets with at least one moon can have one – planets with no moons (Mercury and Venus) cannot.

P1 Chapter 4 Pinchpoint

Pinchpoint question

Correct answer: **B** – "Our Solar System is made up of a star with planets that orbit it." (A is wrong: the Earth orbits the Sun, not the other way round. C is wrong: the correct order smallest to largest is Earth, Sun, Solar System, Milky Way, Universe. D is wrong: the Milky Way is not inside our Solar System – the Solar System is inside the Milky Way.)

Follow-up activity B (matches the correct answer, B)

Option 3 is correct: Moon – 30× Earth diameter away; Sun – about 400× further away than the Moon; Proxima Centauri – about 300 000× further away than the Sun; Andromeda – about 500 000× further away than Proxima Centauri. (These follow directly from the light-time distances given earlier in the booklet: Moon ≈ 1 light-second, Sun ≈ 8 light-minutes, Proxima Centauri ≈ 4 light-years, Andromeda ≈ 2 million light-years.)